STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 2: Motion in a Straight Line

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

v=dx/dt.

Suggested allocation by section: 1-mark questions earn one mark for the correct fact, formula or unit; 2-mark questions normally award one mark for the principle/formula and one for the explanation or result; 3-mark questions award credit for the relation, working/reasoning and result with unit; 4-mark questions award credit for the principle, equation, working and final answer/interpretation. Accept equivalent correct methods.

Q2 1 mark

Velocity.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q3 1 mark

Zero.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q4 1 mark

No.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q5 1 mark

m s-^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q6 1 mark

Zero.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q7 1 mark

v=u+at.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q8 1 mark

Signed displacement.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q9 2 marks

5 m/s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q10 2 marks

2 m/s^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q11 2 marks

5 s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q12 2 marks

25 m/s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q13 2 marks

11 m; 5 m east.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q14 2 marks

12 m/s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q15 3 marks

v=16 m/s; s=55 m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q16 3 marks

a=-4 m/s^2; s=50 m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q17 3 marks

a=2 m/s^2; displacement=48 m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q18 3 marks

v=24 m/s; s=144 m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q19 4 marks

5 s; 62.5 m; acceleration opposes motion.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q20 4 marks

v=3-2t; a=-2 m/s^2; t=1.5 s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.